SIDHO KANHO BIRSHA UNIVERSITY

A project submitted in partial fulfilment of the requirement for the Honours Degree of Bachelor of Computer Science

ONLINE VOTING SYSTEM – 'E-VOTING'

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Dated : __/__/___

Place : __________

SELF CERTIFICATE

This is to certify that the dissertation/project report entitled CREATING ONlINE VOTING PORTAL is done by me and it is an authentic work carried out for the partial fulfilment of the requirements for the award of the degree of BSC under the guidance of Mr. Ranjan Kumar Mishra of Computer Science Department of JK College, Purulia.

I also certify that that I am aware of the "B.SC. PROJECT & PROJECT REPORT STANDARD 2023, SIDHO KANHO BIRSHA UNIVERSITY" issued by Sidho Kanho Birsha University and this project report is based on that standard. The matter embodied in this project work has not been submitted earlier for award of any degree or diploma to the best of my knowledge and belief.

………………………………………………………………………………

Roll : _______ No: ______

Computer Science Department,

JK College, Purulia

CERTIFICATE FROM

This is to certify that this project entitled "ONLINE VOTING SYSTEM– 'e-Voting'", submitted in partial fulfilment of the degree of B.Sc. in Computer Science from JK College of Sidho Kanho Birsha University, done by Soumyajit Kar, Rajib Mahato, Anwesha Bannerjee and Thakurdas Laya, is an authentic work carried by them, under the guidance of Mr. Ranjan Kumar Mishra.

____

Signature of Superviser

Examined by :

____

Signature of Examiner

INDEX

TOPICS **PAGE NO.**

OBJECTIVE OF THE PROJECT

- This Web Application, named **e-Voting**, provides a convenient voting for eligible voters who may have difficulty in accessing traditional polling stations due to mobility issues, distance, or other reasons.
- It reduces vote times, administrative burdens, and costs associated with traditional voting methods.
- To encouraging higher voter turnout by providing a convenient and accessible voting option that fits into voters' busy schedules.

PURPOSE AND SCOPE OF THE PROJECT

PURPOSE

It is an easy process to vote, and a convenient, accessible and secure way for eligible individuals to cast their votes using internet-enabled devices. To increase the voter turnout, streamline the voting process, and potentially reduce costs associated with traditional paper-based voting systems, additionally, it can enhance accessibility for voters with disabilities.

SCOPE

- This includes the development and maintenance of secure online platforms or applications that facilitate the voting process. It involves considerations such as user authentication, encryption, data storage, and system resilience against cyber threats.
- It can be used anywhere, any time as it is a web-based application (user location doesn't matter).
- It is accessible to all eligible voters, including those with disabilities or limited access to technology. This may involve providing alternative voting methods or accommodations to ensure inclusivity.

CATEGORY OF THE PROJECT

- It is web-based application. So, the users are able to access it through web browser.
- It is multi user. We developed this project in this way so that multiple users can access the website simultaneously and can submit their votes at the same time.

DEFINITION OF THE PROBLEM

In the era defined by digital transformation, traditional voting methods are evolving to meet the demands of the modern society. The advent of online voting system presents a promising solution to enhance the accessibility, efficiency and integrity of the electoral process.

The aim of our project(e-Voting) is to develop a comprehensive online voting system that revolutionizes the way elections are conducted. By leveraging cutting-edge technology, robust security measures, and user-friendly interfaces.

This project endeavours to empower voters, streamline administrative processes, and upload the democratic principles of transparency and accountability. Through this project we endeavour to contribute to the advancement of the democratic practices and facilitate greater civic engagement in the digital age.

'Sign In' module helps the user to login to the site. For that he/she must put the username and password correctly. The login provision in this page helps the already registered user to directly access the site and there is a link for registration to a user who is new to this site. The 'About' module describes the purpose of the application and how it's useful and shows user the formats of the votes for Freshers and Experienced, from where and how they can submit their votes.

The 'Contact' module allows the user to send any queries.

SYSTEM REQIREMENT SPECIFICATION

This Online Voting System – **'e-Voting'** web application is developed, so that the user can register online and vote according to their wish, from anywhere at any time.

PURPOSE

You never get a second chance for first impression, and the **e-Voting** web application is made to do this job of creating a good and strong first impression.

The purpose of this application is to help user by providing an easy internet enabled voting process, so that the voters can vote instantly without any hazard. The main objective is to make the voting process effective, efficient, secure and accessible from anywhere at any time.

SCOPE

This web application will be extremely beneficial for the voters to vote from anywhere at any time, especially for those who are physically disabled.

- Voters can register themselves and also login if already has registered.
- Voters can select the party to which it will vote for.
- Voters can see the results after the ending of the voting process.

FUNCTIONAL REQUIREMENTS

User module :

- Home
- About
- Contact
- Login

User End :

- **Home:** It facilitates the students\users to vote after log in.
- **About:** About the Online Voting System.
- **Register:** Here, a Student/User can register for voting.
- **Contact:** User can contact with us i.e. send any queries.
- **Login:** The user can login by giving his/her email id and password.
- **Admin Login:** Admin can login and manage voter registration.

ADMINISTRATOR :

Admin can view the result, students who have logged in and members as well as admin can change the student details also.

NON-FUNCTIONAL REQUIREMENT

- **Database :** A Database management system that is available free of cost in the public domain should be used.
- **Platform :** Both Windows or Unix versions need to be developed.
- **Web-support :** It should be possible to invoke the query book functionality from any place by using a web browser.
- **Observation :** Since there are many requirements, the requirements have been organized into sections. This would not only enhance the readability of the document, but would help in design.

DETAILS OF HARDWARE & SOFTWARE USED FOR DEVELOPMENT

Hardware used for Development

- **Processor :**

  Ryzen Core(i5) CPU (recommended)

- **RAM :**

  8.00 GB

- **HARD DRIVE :**

  512 GB SSD of available space required for installing Macromedia Dreamweaver 8, Adobe Photoshop, WAMP Control Panel.

- Modem and Internet Connection

Software used for Development

- **Operating System :**

  WINDOWS 11

- **Internet Browser :** Any of the following –
- ○ INTERNET EXPLORER 5.5 or above (preferred)

  ○ GOOGLE CHROME

- HTML, CSS, JavaScript for client-side scripting
- PHP for server-side scripting
- MySQL as Database System
- XAMPP Control Panel

THEORETICAL BACKGROUND OF THE SOFTWARE USED

HTML (HYPERTEXT MARKUP LANGUAGE)

HTML or Hypertext Markup Language is the underlying language of the web. It's used to create the structure and contents of the web pages. HTML is the widely-used, free, and efficient. This client-side language is often assisted by technologies such as CSS (Cascading Style Sheets) and scripting languages such as JavaScript. Also, server side scripting language like PHP (Hypertext Preprocessor) can be embedded in HTML directly. HTML is a standard markup language, where markup means, it uses tags to define the structure of a web page. For example, <p>….</p> tag defines a new paragraph, <h1>….</h1> defines heading, <img> tag defines an image etc.

HTML can embed programs written in a scripting language such as JavaScript, which affects the behaviour and content of web pages. The inclusion of CSS defines the look and layout of the content. Whereas embedded PHP in HTML allows developers to create dynamic content and interact with database.

ADVANTAGES OF HTML :-

- It's relatively simple language to learn, even for beginners, as the syntax is straightforward and easy.
- As HTML is free and open-source language, everyone can use and modify it. It's an affordable option for web development.
- It is widely supported i.e. it can be run in all major web browsers.
- HTML files are typically very small and lightweight, so it can load quickly in browser and provides good experience.
- It's an extensible and search engine friendly.
- HTML can be easily integrated with other languages.

CSS (CASCADING STYLE SHEETS)

CSS or Cascading Style Sheets is a language used for describing the presentation of a document written in a markup language such as HTML or XML. The term 'cascading' comes from specified priority scheme to determine which style rule applies if more than one rule matches a particular element. It's basically used to control the appearance of web pages. The fonts, colour of the background and text can be controlled with CSS. Layout of web pages, including the position, size, and spacing of elements can also be controlled using CSS.

ADVANTAGES OF CSS :-

- Powerful tool for controlling the appearance of web pages. It can be used to control the font, colour, size, layout, and other aspects of the appearance of web pages, which allows web developers to create visually appealing and user-friendly websites.
- It’s relatively easy to learn and use as syntax is simple.
  - CSS is widely supported by all major web browsers.
- CSS is extensible and user friendly
- It can be used to create responsive web designs which can adapt to the size of the device that are being viewed on.
- CSS can be used to create animation and transition.

JAVASCRIPT

JavaScript is a lightweight object-oriented programming language, which is used by several websites by scripting the web pages. It’s an interpreted full-fledged programming language, that enables dynamic interactivity on website, when applied to an HTML document. JavaScript is a single threaded language, which means only one piece of code can be executed at a time.

ADVANTAGES OF JAVASCRIPT: -

- It's easy to learn as the syntax and structure is similar to C language.
- All popular web browsers support JavaScript, as they provide built in execution environment.
- It's supportable in several OS (including Windows, Linux and Mac-OS).
- Less server interaction is possible, as client side reflection doesn't falls on the server side.
- Provides good control to user over the web browser.
- JavaScript is very versatile as it can be used for a wide variety of tasks including – adding interactivity to web pages, creating animation effects, validating user input, communicating with the server etc.

PHP (HYPERTEXT PREPROCESSOR)

PHP or Hypertext Preprocessor is a powerful server-side scripting language for creating dynamic and interactive websites. It's a widely, free, and efficient alternative to competitors such as Microsoft's ASP. PHP is perfectly suited for Web development and can be embedded directly into the HTML code. It's a general-purpose language, originally designed for web development to produce dynamic web pages. For this purpose, PHP code is embedded into the HTML source document and interpreted by a web server with a PHP processor module, which generates the web page document. It also has evolved to include a command-line interface capability and can be used in standalone graphical applications. Embedded PHP in HTML allows developers to create dynamic content and interact with the database.

ADVANTAGES OF PHP:-

- It's open source and free to use, so anyone can easily use PHP without having to pay any licensing fees.
- PHP is widely supported by all major web hosting providers and most web browsers.
- It's easy to learn as it has simple syntax.
- PHP is very fast and efficient, which makes it ideal for websites that need to load quickly. Along with that, it's also very efficient in terms of memory.
- It's a very flexible and scalable language. So, it's ideal for variety of web development projects from simple to complex.
- It has built-in support for variety of popular databases.

MYSQL

MySQL is the world's most used relational database management system (RDBMS) that runs as a server providing multi-user access to a number of databases. The SQL phrase stands for ‘Structured Query Language’. The MySQL development project has made its source code available under the terms of the GNU General Public License, as well as under a variety of proprietary agreements. Free-software-open-source projects that require a full-featured database management system often use MySQL. MySQL is a popular choice of database for use in web applications. MySQL is a popular choice of database for use in web applications, and is a central component of the widely used LAMP open-source web application software stack—LAMP is an acronym for "Linux, Apache, MySQL, Perl/PHP/Python". It’s an open-source database management system and is used in some of the most frequently visited websites on the Internet.

ADVANTAGES OF MYSQL: -

- MySQL is an open-source database, so anyone can download and use it for free. This makes it very cost-effective option for business and organization.
- As MySQL has simple syntax, it’s easy to learn.
- It has very scalable database, so it can be easily expanded to handle increased traffic or data volumes.
- MySQL is a very popular database and there’s a large community of developers who support it. So, there’re many

resources available to help users learn and use it properly.

 It’s fast and efficient, which makes it ideal for websites that need to load quickly. Along with that, it’s also vey efficient in terms of memory usage.

 MySQL is extensible database.

DETAILED LIFE CYCLE OF THE PROJECT

DFD (DATA FLOW DIAGRAM)

A Data Flow Diagram or DFD is a graphical representation of the "flow" of data through an information system. A data flow diagram can also be used for the visualization of data processing (structured design).

ELEMENTS OF A DFD

There are 4 key elements in a Data Flow diagram –

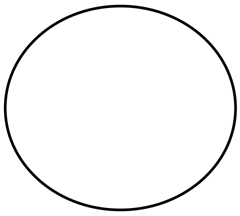
- Processes
- Data Flows
- Data Stores
- External entities

Each element is drawn differently. Now we'll discuss about all these elements one by one.

1. **Process entity :** The Process entity identifies a process taking place; it must have at least one input and output. A process with no input is known as a "miracle process" and one with no output is a "black hole process". Each process has the following –
    - A Number
    - A Name (verb phrase)

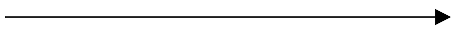
- A Description
- At least one input
- At least one output

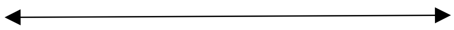
Process entity is drawn as following figure –



2. **<u>Data Flow Entity</u> :** The Data Flow entity identifies the flow of data between processes, data stores & external entities. A data flow cannot connect an external entity to a data source; at least one connection must be with a process. There are also "physical" flows, i.e. those that use a physical medium. Each data flow has the following components – • A Name (Noun)
    - A Description
    - One or more connections to a process. Data flow

    is shown as following figure –

    Unidirectional : 

    Bidirectional : 

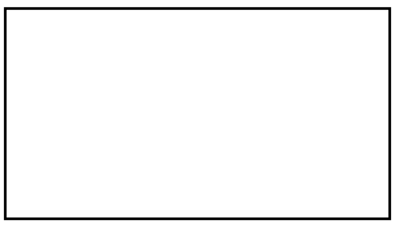
3. **<u>Data Store Entity</u> :** The Data Store entity identifies stores of data, both manual and electronic. Each data store has the following –
    - A Number

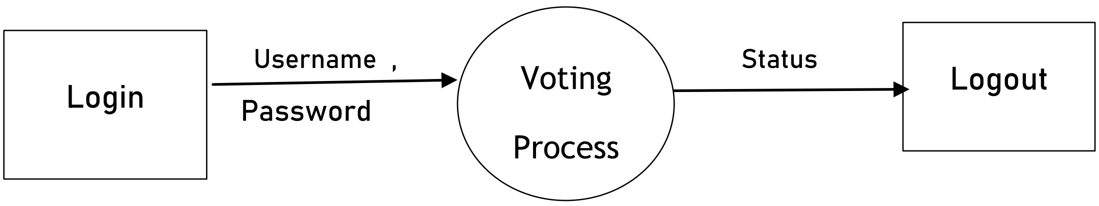
- A Name
- A Description
- One or more input data flows
- One or more output data flows

Data store entity is shown as following figure –

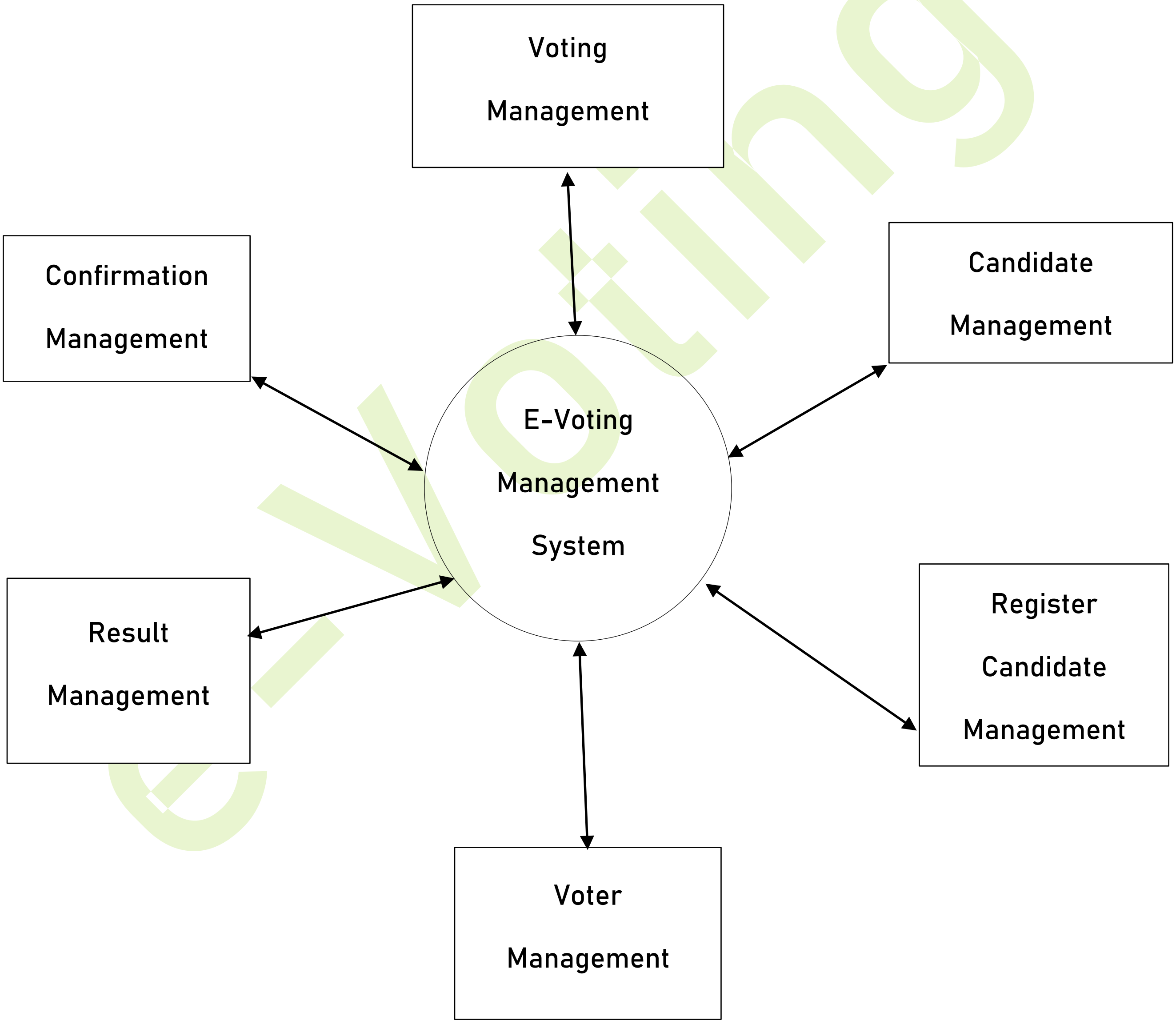
4. **<u>External Entity</u> :** The External Entity identifies external entities which interacts with the system, usually clients but can be within the same organization. Each external entity has the following –
   - A Name (Noun)
   - A Description

External entity is drawn as the following figure –





Zero Level DFD of Online Voting System



Level One DFD of Online Voting System

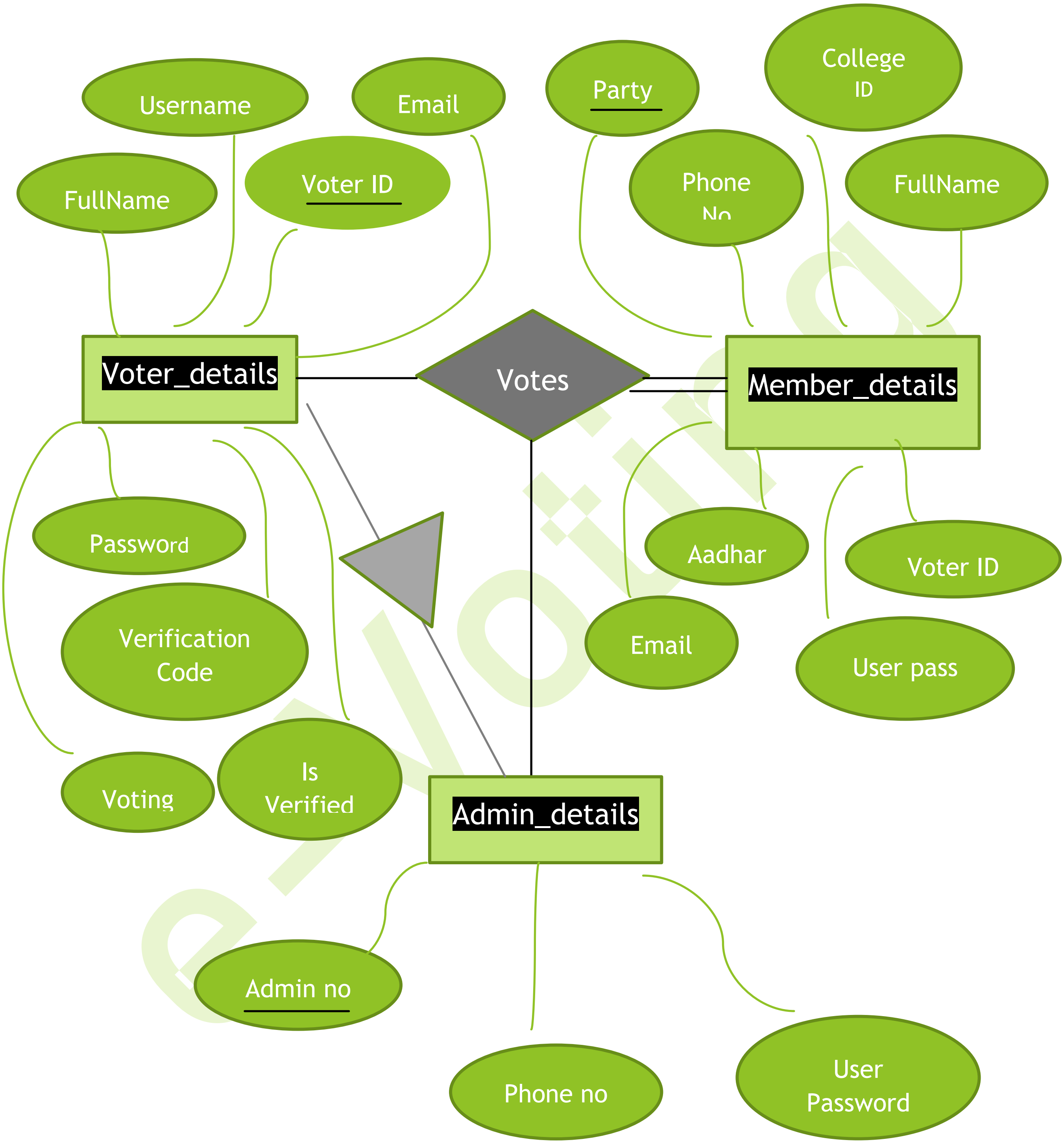
Section – 1: Relational Schemas of the table / Entity sets

i. **voter_details(**full_name, username, aadhar, email, password, verification_code, is_verified, resulttoken, resulttokenexpire, voting, party**)**

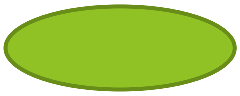
ii. **member_details(**party, college_id, phone_no, fullName, aadharNum, voterNum, email, user_pass, voters**)**

iii. **admin_details(**admin_no, phone_no, user_pass**)**

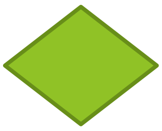
Section – 2: ER-Diagram for 'E-voting'

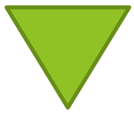


Section-3: Symbol used for this

1.  ----- Attributes of an entity set.

2.  ----- Represent an entity set

3.  ------ Represent the relationship

4.  ----- Specifies Specialization

Software Engineering Process Involved

There are various models in the study of software engineering, which provide structured approach to software development and help manage the cost, risk, schedule of the project. These models basically give a pathway to make a high-quality application easily. Our project i.e. Online Voting System– **'e-Voting'** is based on Agile Model.

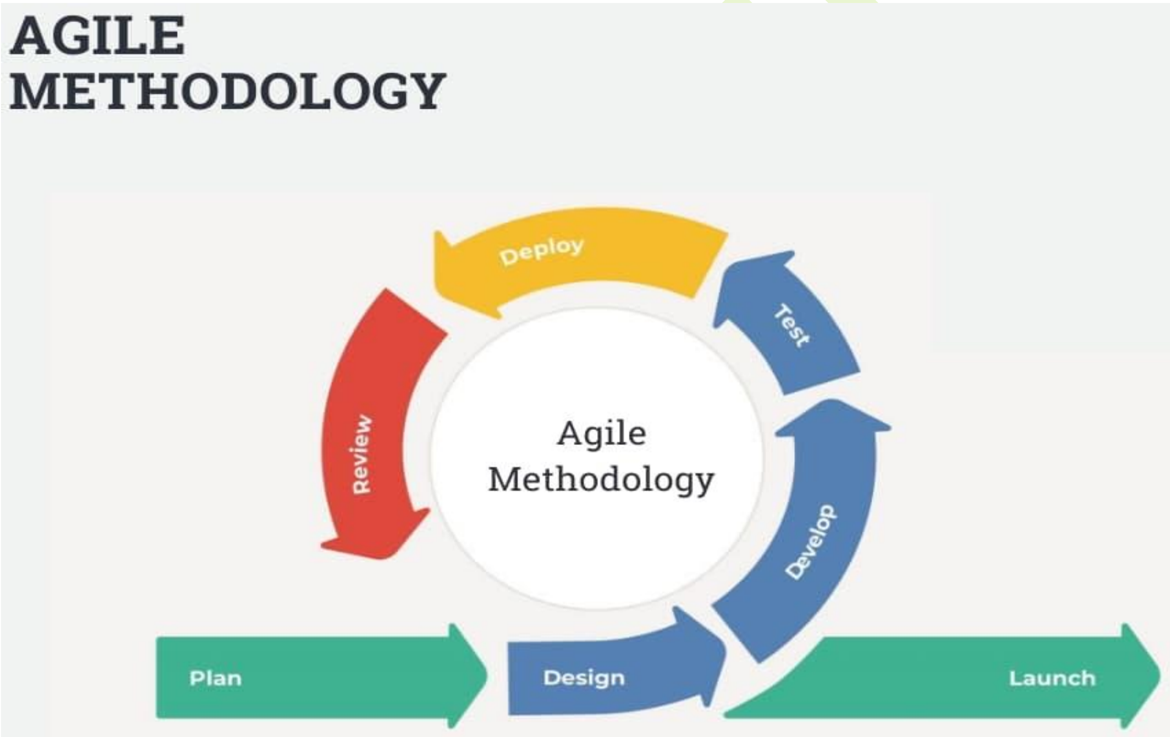
The Agile methodology is a project management and software development approach that emphasizes on flexibility, collaboration, and customer-centricity. It follows the iterative as well as incremental approach that emphasizes the importance of delivering of working product quickly.

Why Agile Model for Our Project?

The Agile model is suitable for developing our project i.e. Online Voting System("e-Voting") as it allows for iterative development and frequent releases of software. This means that we can continuously enhance and refine the online voting system based on feedback and changing requirements. Agile methodology can help us to ensure that the voting system is robust and secure and this model emphasizes on flexibility and adaptability.

PROCEDURE OF AGILE MODEL

The agile model follows the set of procedure or practices that help teams effectively manage and deliver software development projects. Here are the key procedures typically followed in agile methodology shown in following figure.



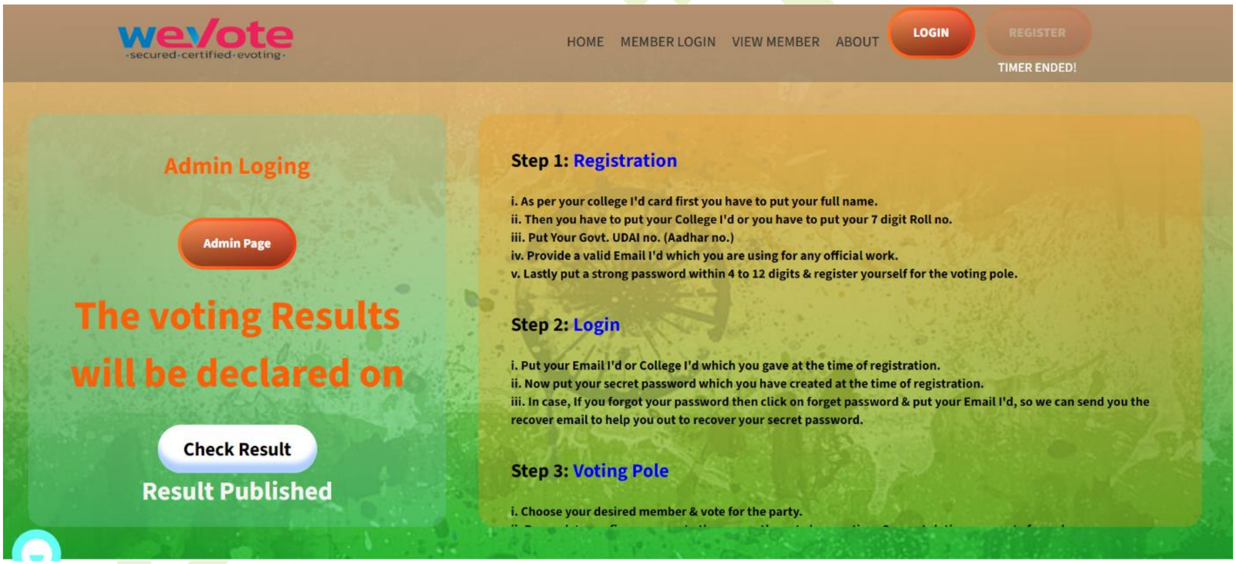
There are 7 phases followed by agile model and these are:

PLANNING

- At this phase we have discussed about the project, what is supposed to achieve and the problem it solves, then we have planned according to the requirements.
- Tasks are defined, estimated, and assigned to our team members based on their skills and availability.

DESIGN

In this phase we have designed a dummy interface of our project, shown below:



DEVELOPE

- The development team works on implementing the selected features or user stories.
- Daily stand-up meetings (daily scrums) are held to discuss progress, identify and remove obstacles, and ensure alignment within the team.

- Continuous collaboration between developers, testers, designers, and other stakeholders ensures that the work progresses smoothly.

CODE

Index Page

```
<?php
// Including the connection file
require ('connection.php');

// Start a session
session_start();

// Check if the form is submitted
if ($_SERVER["REQUEST_METHOD"] == "POST") {
    // Check if login credentials are correct (you need to implement this logic)
    // Assuming login is successful
    $username = $_POST['username']; // Assuming username is used for login
    $_SESSION['username'] = $username; // Store username in session
    $_SESSION['logged_in'] = true; // Set logged_in to true

    // Redirect to the voting page
    header("Location: voting_page.php");
    exit();
}
?>
<!DOCTYPE html>
<html lang="en">

<head>
    <!-- Meta tags -->
    <meta charset="UTF-8">
    <meta http-equiv="X-UA-Compatible" content="IE=edge">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Online Voting System</title>
    <!-- CSS file -->
    <link rel="stylesheet" href="style.css">
    <!-- Inline CSS style for timer -->
    <style>
        #timer {
            color: whitesmoke;
            font-size: xx-large;
            font-weight: bold;
        }

        .button-73.hidden {
            display: none;
        }
    </style>
```

```
    <!-- JavaScript for disabling right-click -->
    <script language="javascript">
      function noClick() {
        if (event.button == 2) {
          alert("you can not click on this page")
        }
      }
      document.onmousedown = noClick
    </script>
  </head>

  <body>
    <div class="container">
      <!-- Header section -->
      <header>
        <!-- Logo -->
        <div class="left">
          <img src="eci.png" alt="Logo">
        </div>

        <!-- Hamburger menu -->
        <div class="hamburger" onclick="toggleMenu()">
          <div></div>
          <div></div>
          <div></div>
        </div>

        <!-- Navigation links -->
        <nav id="navLinks">
          <?php if (!(isset($_SESSION['logged_in']) && $_SESSION['logged_in'] == true)) { ?>
            <!-- If user is not logged in -->
            <a href="index.php">HOME</a>
            <a href="memberlogin.php">Member Login</a>
            <a href="memberdetailes.php">View Member</a>
            <a href="#">ABOUT</a>
          <?php } ?>
          <?php if (isset($_SESSION['logged_in']) && $_SESSION['logged_in'] == true) { ?>
            <!-- If user is logged in -->
            <h2 class="tages">Online Voting Procedure</h2>
            <?php echo "<script>document.location.href='voting_page.php'</script>"; ?>
          <?php } else { ?>
            <!-- If user is not logged in, show login and registration buttons -->
            <div class="button">
              <div class="sign-in-up">
                <button class="button-78" id="login-btn" type="button"
                  onclick="popup('login-popup')">LOGIN</button>
                <div id="login-timer"></div>
              </div>
              <div class="sign-in-up">
                <button class="button-78" id="register-btn" type="button"
                  onclick="popup('register-popup')">REGISTER</button>
                <div id="register-timer"></div>
              </div>
            </div>
          <?php } ?>
        </nav>
      </header>

      <!-- Main content -->
      <div class="box">
        <!-- Box 1: Information about voting results and timer -->
        <div class="box1">
```

```
<p><br>
<h2>Admin Login</h2> <br>
</p>
<div class="button">
  <div class="sign-in-up">
    <button class="button-78" id="view" type="button"
      onclick="location.href='/voting/addministrator/admin.php'">Admin Page</button><br>
  </div>
</div>
<p><br>
<h1>The voting Results will be declared on</h1> <br>
</p>
<!-- Button to check results (hidden until the time) -->
<button id="check-result-btn" class="button-73 hidden" style="vertical-align:middle"
  onclick="gotoResult()"><span>Check Result</span></button>
<!-- Timer for voting results declaration -->
<div id="timer"></div>
</div>

<!-- Box 2: Steps for voting procedure -->
<div class="box2">
  <p>
  <h2><u>Step 1: </u><a href="registration.png">Registration</a></h2><br>
  <h4>
    i. As per your college I'd card first you have to put your full name.<br>
    ii. Then you have to put your College I'd or you have to put your 7 digit Roll no. <br>
    iii. Put Your Govt. UDAI no. (Aadhar no.)<br>
    iv. Provide a valid Email I'd which you are using for any official work.<br>
    v. Lastly put a strong password within 4 to 12 digits & register yourself for the voting pole.<br>
  </h4>
  <h2><u><br>Step 2: </u><a href="loging.png"> Login</a></h2><br>
  <h4>
    i. Put your Email I'd or College I'd which you gave at the time of registration.<br>
    ii. Now put your secret password which you have created at the time of registration.<br>
    iii. In case, If you forgot your password then click on forget password & put your Email I'd, so we
    can send you the recover email to help you out to recover your secret password.<br>
  </h4>
  <h2><u><br>Step 3:</u><a href="votes.jpg"> Voting Pole<a></h2><br>
  <h4>
    i. Choose your desired member & vote for the party.<br>
    ii. Press ok to confirm your party then press the vote here option. Congratulations you vote for a
    change.<br>
    iii. In case, if you don't want to vote for anyone then directly press the vote here button.<br>
  </h4>
  </p>
</div>
</div>

<!-- Displaying welcome message if user is logged in -->
<?php
if (isset($_SESSION['logged_in']) && $_SESSION['logged_in'] == true) {
  echo "<h1 style='text-align: center;margin-top:200px'>welcome to the voting page - $_SESSION[username]</h1>";
}
?>
</div>

<!-- Popups for login, registration, and password reset -->
<div class="popup-container" id="login-popup">
  <div class="popup">
    <form method="POST" action="login_register.php">
      <h2>
        <span>USER LOGIN</span>
```

```
            <button type="reset" onclick="popup('login-popup')">X</button>
        </h2>
        <input type="text" placeholder="E-mail or college_id" name="email_username">
        <input type="password" placeholder="Password" name="password">
        <button type="submit" class="login-btn" name="login">LOGIN</button>
    </form>
    <div class="forgot-btn">
        <button type="button" onclick="forgotPopup()">Forgot Password?</button>
    </div>
  </div>
</div>

<div class="popup-container" id="register-popup">
  <div class="register popup">
    <form method="POST" action="login_register.php">
        <h2>
            <span>USER REGISTER</span>
            <button type="reset" onclick="popup('register-popup')">X</button>
        </h2>
        <input type="text" placeholder="Full Name" name="fullname" require>
        <input type="number" placeholder="College id" name="username" require>
        <input type="number" placeholder="Aadhar Number" name="aadherno" require>
        <input type="email" placeholder="E-mail" name="email" require>
        <input type="password" placeholder="Password" name="password" require>
        <input type="password" placeholder="Confirm Password" name="cnfpassword" require>
        <button type="submit" class="register-btn" name="register">REGISTER</button>
    </form>
  </div>
</div>

<div class="popup-container" id="forgot-popup">
  <div class="forgot popup">
    <form method="POST" action="forgotpassword.php">
        <h2>
            <span>Reset Password</span>
            <button type="reset" onclick="popup('forgot-popup')">X</button>
        </h2>
        <input type="email" placeholder="E-mail" name="email">
        <button type="submit" class="reset-btn" name="send-reset-link">Send Link</button>
    </form>
  </div>
</div>

<!-- JavaScript for hamburger menu, timer, and popups -->
<script>
  // JavaScript for hamburger menu
  function toggleMenu() {
    var navLinks = document.getElementById("navLinks");
    var hamburger = document.querySelector('.hamburger');
    if (navLinks.style.display === "flex") {
        navLinks.style.display = "none";
        hamburger.classList.remove('active');
    } else {
        navLinks.style.display = "flex";
        hamburger.classList.add('active');
    }
  }

  // Call toggleMenu on initial page load
  window.onload = function () {
    window.onload = startRegisterTimer;
    var screenWidth = window.innerWidth;
```

```
        if (screenWidth <= 768) {
          document.getElementById("navLinks").style.display = "none";
        }
      }

      // Call toggleMenu on window resize
      window.onresize = function () {
        var screenWidth = window.innerWidth;
        if (screenWidth > 768) {
          document.getElementById("navLinks").style.display = "flex";
        } else {
          document.getElementById("navLinks").style.display = "none";
        }
      }
    </script>

    <!-- External script for collecting data -->
    <script>(function (w, d) { w.CollectId = "6638e5a13e99425e992d292d"; var h = d.head ||
d.getElementsByTagName("head")[0]; var s = d.createElement("script"); s.setAttribute("type", "text/javascript"); s.async =
true; s.setAttribute("src", "https://collectcdn.com/launcher.js"); h.appendChild(s); })(window, document);</script>

    <!-- JavaScript for registration timer & loging-->
    <script>
      function startTimers() {
        const loginButton = document.getElementById('login-btn');
        const loginTimer = document.getElementById('login-timer');
        const registerButton = document.getElementById('register-btn');
        const registerTimer = document.getElementById('register-timer');

        const now = new Date().getTime();
        const registerEndDate = new Date('june 25, 2024 14:18:00').getTime(); // Set your registration end date
        const loginEndDate = new Date('July 02, 2024 00:00:00').getTime(); // Set login end date

        function updateTimers() {
          const now = new Date().getTime();

          // Update registration timer
          const registerTimeLeft = registerEndDate - now;
          if (registerTimeLeft <= 0) {
            registerButton.disabled = true;
            registerTimer.innerHTML = "Registration End!";
            loginButton.disabled = false; // Enable the login button
          } else {
            const registerDays = Math.floor(registerTimeLeft / (1000 * 60 * 60 * 24));
            const registerHours = Math.floor((registerTimeLeft % (1000 * 60 * 60 * 24)) / (1000 * 60 * 60));
            const registerMinutes = Math.floor((registerTimeLeft % (1000 * 60 * 60)) / (1000 * 60));
            const registerSeconds = Math.floor((registerTimeLeft % (1000 * 60)) / 1000);
            registerTimer.innerHTML = `${registerDays}d ${registerHours}h ${registerMinutes}m ${registerSeconds}s`;
          }

          // Update login timer
          if (!registerButton.disabled) {
            loginButton.disabled = true; // Ensure login button is disabled while registration is open
          } else {
            const loginTimeLeft = loginEndDate - now;
            if (loginTimeLeft <= 0) {
              loginButton.disabled = true;
              loginTimer.innerHTML = "Vote End!";
            } else {
              const loginDays = Math.floor(loginTimeLeft / (1000 * 60 * 60 * 24));
              const loginHours = Math.floor((loginTimeLeft % (1000 * 60 * 60 * 24)) / (1000 * 60 * 60));
              const loginMinutes = Math.floor((loginTimeLeft % (1000 * 60 * 60)) / (1000 * 60));
```

```
                const loginSeconds = Math.floor((loginTimeLeft % (1000 * 60)) / 1000);
                loginTimer.innerHTML = `${loginDays}d ${loginHours}h ${loginMinutes}m ${loginSeconds}s`;
            }
        }
    }

    // Initially disable the login button
    loginButton.disabled = true;

    // Start the timer updates
    setInterval(updateTimers, 1000);
  }

  window.onload = startTimers;

</script>
<!-- JavaScript for voting timer and check result -->
<script>
  function gotoResult() {
    window.location.href = "result_page.html"; // Replace with the actual result page URL
  }

  function updateTimer(remainingTime) {
    const timerElement = document.getElementById('timer');
    const buttonElement = document.getElementById('check-result-btn');
    if (remainingTime <= 0) {
      timerElement.textContent = 'Result Published';
      buttonElement.classList.remove('hidden'); // Show the button
    } else {
      const days = Math.floor(remainingTime / (3600 * 24));
      const hours = Math.floor((remainingTime % (3600 * 24)) / 3600);
      const minutes = Math.floor((remainingTime % 3600) / 60);
      const seconds = remainingTime % 60;
      timerElement.textContent = `${days}d ${hours}h ${minutes}m ${seconds}s`;
      buttonElement.classList.add('hidden'); // Hide the button
    }
  }

  function checkButtonStatus() {
    fetch('check_timer.php')
      .then(response => response.json())
      .then(data => {
        updateTimer(data.remainingTime);
      })
      .catch(error => console.error('Error:', error));
  }

  // Check button status and update timer every second
  setInterval(checkButtonStatus, 1000);

  // Initial check when the page loads
  checkButtonStatus();
</script>
<!-- JavaScript for handling popups -->
<script>
  function popup(popup_name) {
    get_popup = document.getElementById(popup_name);
    if (get_popup.style.display == "flex") {
      get_popup.style.display = "none";
    } else {
      get_popup.style.display = "flex";
    }
```

```
        }

        function forgotPopup() {
            document.getElementById('login-popup').style.display = "none";
            document.getElementById('forgot-popup').style.display = "flex";
        }

        function gotoResult() {
            window.location.href = "result.php";
        }
    </script>
</body>

</html>
```

Voting Page

```
<?php
// Start the session
session_start();

// Check if the user is logged in
if (!isset($_SESSION['logged_in']) || $_SESSION['logged_in'] !== true) {
    // If not logged in, redirect to login page
    header("Location: index.php");
    exit();
}

// Now you can access the user's data using $_SESSION['username']
$username = $_SESSION['username'];
?>

<!-- <?php
require ('connection.php');

// Check if user is logged in
if (!isset($_SESSION['username'])) {
    header("Location: index.php"); // Redirect to login page if not logged in
    exit();
}

// Check if form is submitted
if ($_SERVER["REQUEST_METHOD"] == "POST") {
    if (isset($_POST['selectedPerson'])) {
        $selectedPerson = $_POST['selectedPerson'];

        // Update the vote count for the selected person in the database
        $query = "UPDATE candidates SET votes = votes + 1 WHERE name = ?";
        $stmt = $conn->prepare($query);
        $stmt->bind_param("s", $selectedPerson);
        $stmt->execute();

        // Redirect to the login page after voting
        header("Location: index.php");
        exit();
    }
}
?> -->
<?php

session_unset();
session_destroy();
```

```
header("index.php");
?>

<!DOCTYPE html>
<html>

<head>
   <meta charset="UTF-8">
   <title>online voting system</title>
   <link href="https://unpkg.com/boxicons@2.1.4/css/boxicons.min.css" rel="stylesheet">
   <link rel="stylesheet" href="style3.css">
   <link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.5.2/css/all.min.css"
      integrity="sha512-
SnH5WK+bZxgPHs44uWIX+LLJAJ9/2PkPKZ5QiAj6Ta86w+fsb2TkcmfRyVX3pBnMFcV7oQPJkl9QevSCWr3W6A=="
      crossorigin="anonymous" referrerpolicy="no-referrer" />
       <script language="javascript">
      function noClick()
      {
         if(event.button==2)
         {
            alert("This function is not work!!")
         }
      }
      document.onmousedown=noClick
   </script>
</head>

<body>
   <div class="container">

      <div class="navbar">
         <div class="left" onclick="toggle()">
            <img src="eci.png" alt="voting" />
            <h3>College_id:-<?php echo $username; ?></h3>
         </div>
         <div class="right">
            <button>
               <a href="index.php">Logout</a>
            </button>
            <button>
               <i class="fa-regular fa-user"></i>
            </button>
         </div>
      </div>

      <div class="main">
         <form method="POST" action="votingpage.php">

            <div class="person">
               <img src="skpic.png">
               <h1>
                  Soumyajit
               </h1>
               <p><i class='bx bx-briefcase'></i> Party 1</p>
               <input type="radio" name="person" value="Soumyajit" onclick="selectperson('Soumyajit')">

            </div>
            <div class="person">
               <img src="rmpic.png">
               <h1>
                  Rajib
               </h1>
```

```
            <p><i class='bx bx-briefcase'></i> Party 2</p>
            <input type="radio" name="person" value="Rajib" onclick="selectperson('Rajib')">

          </div>
          <div class="person">
            <img src="tdmpic.png">
            <h1>
              Thakurdas
            </h1>
            <p><i class='bx bx-briefcase'></i> Party 3</p>
            <input type="radio" name="person" value="Thakurdas" onclick="selectperson('Thakurdas')">

          </div>
          <div class="person">
            <img src="abpic.png">
            <h1>
              Anwesha
            </h1>
            <p><i class='bx bx-briefcase'></i> Party 4</p>
            <input type="radio" name="person" value="Anwesha" onclick="selectperson('Anwesha')">
          </div>

          <input type="hidden" name="selectedPerson" id="selectedPersonInput" value="">
          <input type="hidden" name="username" value="<?php echo $username; ?>">
          <div id="selectedPerson">Select Your Vote</div>
          <input type="submit" name="complete_vote" value="Vote Here">
        </form>

        <!-- <div class="details" id="profile">
        <img src="placeholder.jpg">
        <div>
          <h1>
            <%=vote %>
          </h1>
          <h1 style="font-size:13px; font-style:italic; font-weight:500;">Experience: <%=exp %> yrs</h1>
        </div>
        <a href="deleteVote.jsp?id=<%=username%>"><i class='bx bx-trash'></i></a>


      </div> -->
        <h1 class="voted" id="voted">Click Vote Here Button </h1>
      </div>
    </div>

    <script>
      var person = "";
      const person_container = document.getElementById("selectedPerson");
      const hiddenInput = document.getElementById("selectedPersonInput");

      function selectperson(name) {
        const response = confirm("Are you sure to vote " + name + " ?");
        if (response === true) {
          var allRadioButtons = document.querySelectorAll('input[type="radio"]');
          allRadioButtons.forEach(function (radioButton) {
            radioButton.checked = false;
          });

          // Check the currently clicked radio button
          var currRadio = document.querySelector('input[type="radio"][name="person"][value="' + name + '"]');
          if (currRadio) {
            currRadio.checked = true;
          }
```

```
            if (person === name) {
                alert(name + ' is already selected !!');
            } else {
                person = name;
                console.log(name);
                person_container.innerHTML = "Selected Person: " + person;

                // Set the value of the hidden input field
                hiddenInput.value = person;
            }
        } else {
            var allRadioButtons = document.querySelectorAll('input[type="radio"]');
            allRadioButtons.forEach(function (radioButton) {
                radioButton.checked = false;
            });
        }
        return person;
    }
  </script>
</body>

</html>
```

Registration & Login

```
<?php
// Including the connection to the database
require ('connection.php');
// Starting a session
session_start();
use PHPMailer\PHPMailer\PHPMailer;
use PHPMailer\PHPMailer\SMTP;
use PHPMailer\PHPMailer\Exception;

// Function to send email verification
function sendMail($email, $v_code)
{
    // Including PHPMailer classes
    require ('PHPMailer/PHPMailer.php');
    require ('PHPMailer/SMTP.php');
    require ('PHPMailer/Exception.php');

    // Creating a new PHPMailer object
    $mail = new PHPMailer(true);

    try {
        // Configuring SMTP settings
        $mail->isSMTP();
        $mail->Host = 'smtp.gmail.com';
        $mail->SMTPAuth = true;
        $mail->Username = 'pr*************s196@gmail.com';
        $mail->Password =’xxxxxxxxxxxxxxxx';
        $mail->SMTPSecure = 'ssl';
        $mail->Port = 465;

        // Setting email content
        $mail->setFrom('projectworks196@gmail.com', 'Online voting system');
        $mail->addAddress($email);
        $mail->isHTML(true);
```

```
        $mail->Subject = 'Email verification from online voting system';
        $mail->Body = "<h2>Thanks for registration!<br>
                Please verify your mail...<br></h2>
                <a href='http://localhost/voting/verify.php?email=$email&v_code=$v_code'>
                <h1>Verify your mail</h1>
                </a>";

        // Sending the email
        $mail->send();
        return true;
    } catch (Exception $e) {
        // If an error occurs during email sending, return false
        return false;
    }
}

// Handling login form submission
if (isset($_POST['login'])) {
    // Escaping special characters to prevent SQL injection
    $email_username = mysqli_real_escape_string($con, $_POST['email_username']);
    $password = mysqli_real_escape_string($con, $_POST['password']);

    // Query to select user based on email or username
    $query = "SELECT * FROM `registered_users` WHERE `email`='$email_username' OR `username`='$email_username'";
    $result = mysqli_query($con, $query);

    if ($result) {
        if (mysqli_num_rows($result) == 1) {
            $result_fetch = mysqli_fetch_assoc($result);
            if ($result_fetch['is_verified'] == 1) {
                if (password_verify($password, $result_fetch['password'])) {
                    // If password matches, set session variables and redirect to index.php
                    $_SESSION['logged_in'] = true;
                    $_SESSION['username'] = $result_fetch['username'];
                    header("location: index.php");
                    exit;
                } else {
                    // If password is incorrect, show alert and redirect to index.php
                    echo "
                        <script>
                            alert('Incorrect password');
                            window.location.href='index.php';
                        </script>
                    ";
                }
            } else {
                // If email is not verified, show alert and redirect to index.php
                echo "
                    <script>
                        alert('Email not verified');
                        window.location.href='index.php';
                    </script>
                ";
            }
        } else {
            // If no matching user found, show alert and redirect to index.php
            echo "
                <script>
                    alert('College ID Not Registered! & Fill in the blank box');
                    window.location.href='index.php';
                </script>
            ";
```

```
        }
    } else {
        // If query execution fails, show alert and redirect to index.php
        echo "
            <script>
                alert('Cannot run Query');
                window.location.href='index.php';
            </script>
        ";
    }
}

// Handling registration form submission
if (isset($_POST['register'])) {
    // Check if all fields are filled
    if (empty($_POST['fullname']) || empty($_POST['username']) || empty($_POST['aadherno']) || empty($_POST['email']) ||
empty($_POST['password'])) {
        // If any field is empty, show alert and redirect to index.php
        echo "
            <script>
                alert('Please fill in all fields.');
                window.location.href='index.php';
            </script>
        ";
        exit;
    }

    // Check if password and confirm password match
    if ($_POST['password'] !== $_POST['cnfpassword']) {
        // If passwords do not match, show alert and redirect to index.php
        echo "
            <script>
                alert('Both passwords do not match.');
                window.location.href='index.php';
            </script>
        ";
        exit;
    }
    $user_exists = "SELECT * FROM `student_info` WHERE `college_id`='$_POST[username]' AND `aadher_no` =
'$_POST[aadherno]'";
    $results = mysqli_query($con, $user_exists);

    // Check if username or email is already taken
    $user_exist_query = "SELECT * FROM `registered_users` WHERE `username`='$_POST[username]' OR `email` =
'$_POST[email]'";
    $result = mysqli_query($con, $user_exist_query);
    if ($results) {
        if (mysqli_num_rows($results) == 1) {
            if ($result) {
                if (mysqli_num_rows($result) > 0) {
                    $result_fetch = mysqli_fetch_assoc($result);
                    if ($result_fetch['username'] == $_POST['username']) {
                        // If username is already taken, show alert and redirect to index.php
                        echo "
                    <script>
                        alert('$result_fetch[username] - College ID already registered!');
                        window.location.href='index.php';
                    </script>
                ";
                    } else {
                        // If email is already taken, show alert and redirect to index.php
                        echo "
```

```
                <script>
                  alert('$result_fetch[email] - Email ID already registered!');
                  window.location.href='index.php';
                </script>
              ";
                }
              } else {
                // Inserting new user data into the database
                $password = password_hash($_POST['password'], PASSWORD_BCRYPT);
                $v_code = bin2hex(random_bytes(16));
                $query = "INSERT INTO `registered_users`(`full_name`, `username`, `aadher`, `email`, `password`,
`verification_code`, `is_verified`) VALUES
('$_POST[fullname]','$_POST[username]','$_POST[aadherno]','$_POST[email]','$password','$v_code','0')";

                if (mysqli_query($con, $query) && sendMail($_POST['email'], $v_code)) {
                  // If registration is successful, show alert and redirect to index.php
                  echo "
                <script>
                  alert('Registration successful');
                  window.location.href='index.php';
                </script>
              ";
                } else {
                  // If data insertion fails, show alert and redirect to index.php
                  echo "
                <script>
                  alert('Server Down!');
                  window.location.href='index.php';
                </script>
              ";
                }
              }
            } else {
              // If query execution fails, show alert and redirect to index.php
              echo "
            <script>
              alert('Cannot Run Query');
              window.location.href='index.php';
            </script>
            ";
            }
        } else
        {
            echo "
            <script>
              alert('Student is not study in the college');
              window.location.href='index.php';
            </script>
        ";
        }

    }
}
?>
```

Result Page

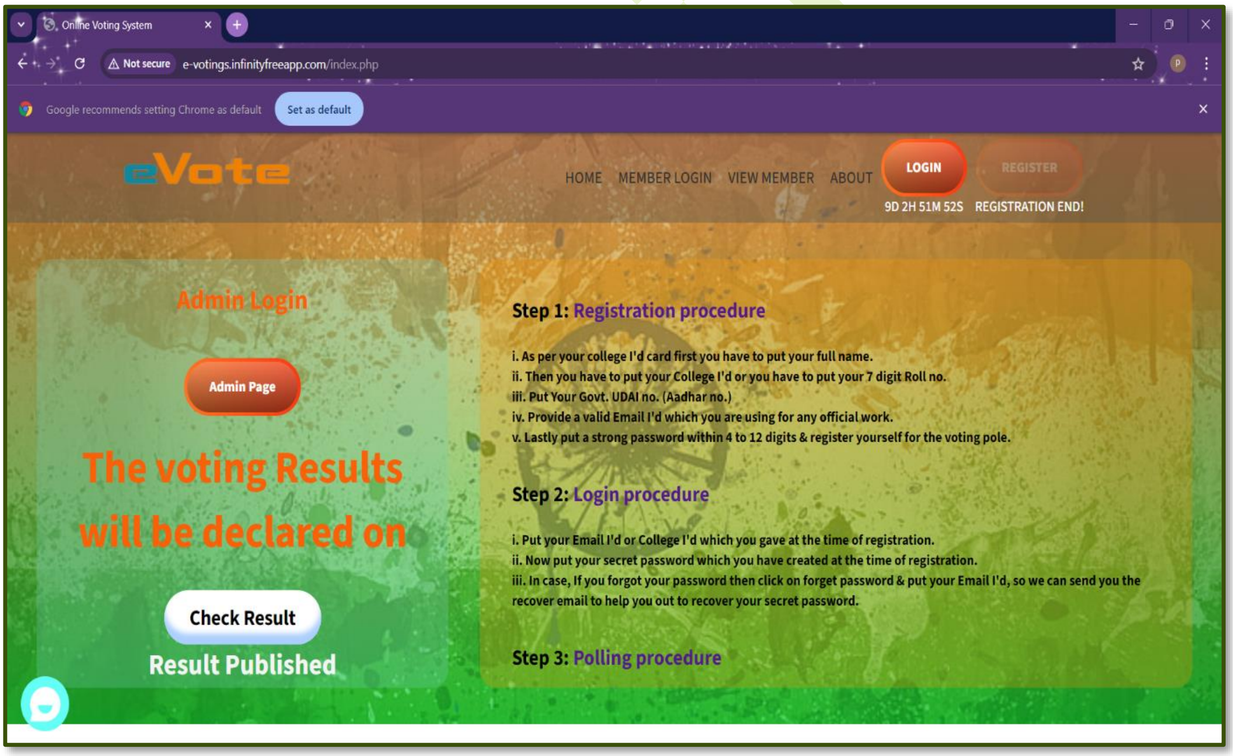
```
<?php
// Including the connection to the database
require('connection.php');
?>
<?php
session_start();
```

```
if (!isset($_SESSION['loggedin']) || $_SESSION['loggedin'] !== true) {
    header("Location: admin.php");
    exit();
}

header("Cache-Control: no-cache, no-store, must-revalidate");
header("Pragma: no-cache");
header("Expires: 0");

// Your protected content here
?>
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Display Data</title>
    <!-- Including external stylesheet for table styling -->
    <link rel="stylesheet" href="tableStyle.css">
    <style>
        h1 {
            text-align: center;
        }
    </style>
    <script language="javascript">
        // JavaScript function to prevent right-clicking
        function noClick()
        {
            if(event.button==2)
            {
                alert("This function is not work!!")
            }
        }
        document.onmousedown=noClick
    </script>
</head>
<body>
<button class="update-button" onclick="location.href='logout.php'">Home</button>
<button class="update-button" onclick="location.href='buttonpage.php'">Back</button>
    <h1>Voting Result</h1>
    <!-- Creating a table to display data -->
    <table>
        <tr>
            <!-- Table headers -->
            <th>Party</th>
            <th>Fullname</th>
            <th>Votes</th>
        </tr>
        <?php
        // Query to select data from the database table
        $sql = "SELECT party, fullname, votes FROM member_details";
        // Executing the query
        $result = $con->query($sql);

        // Checking if there are rows returned from the query
        if ($result->num_rows > 0) {
            // Looping through each row of the result set
            while ($row = $result->fetch_assoc()) {
                // Outputting each row as a table row in HTML
                echo "<tr>";
                echo "<td>" . $row['party'] . "</td>";
```

```
            echo "<td>" . $row['fullname'] . "</td>";
            echo "<td>" . $row['votes'] . "</td>";
            echo "</tr>";
        }
    } else {
        // Outputting a message if no results are found
        echo "<tr><td colspan='3'>No results found</td></tr>";
    }
    ?>
  </table>
</body>
</html>
```

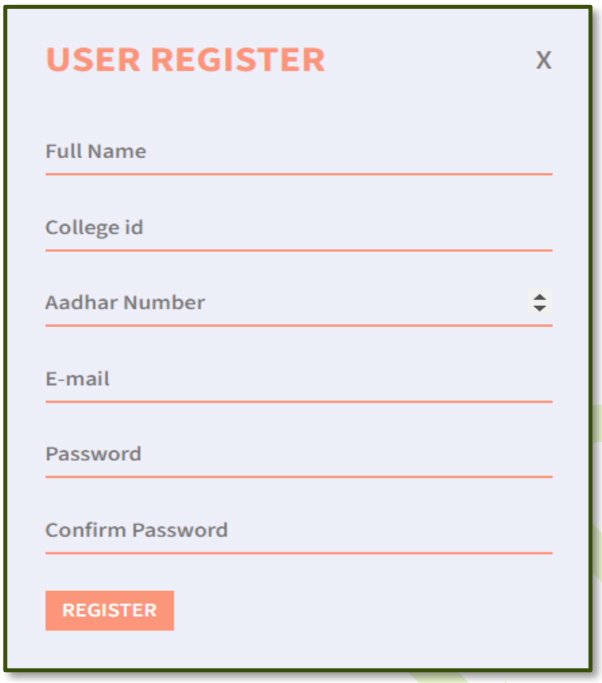
USER'S INTERFACE

Home Page:



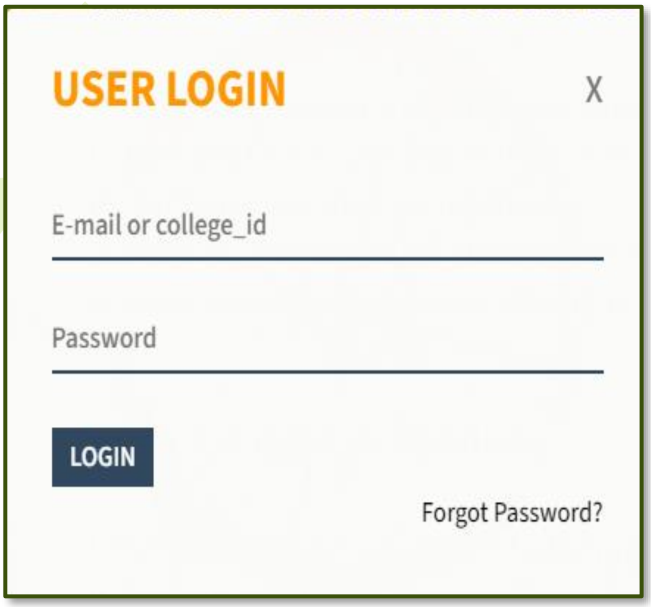
User Registration:

After clicking on register button –



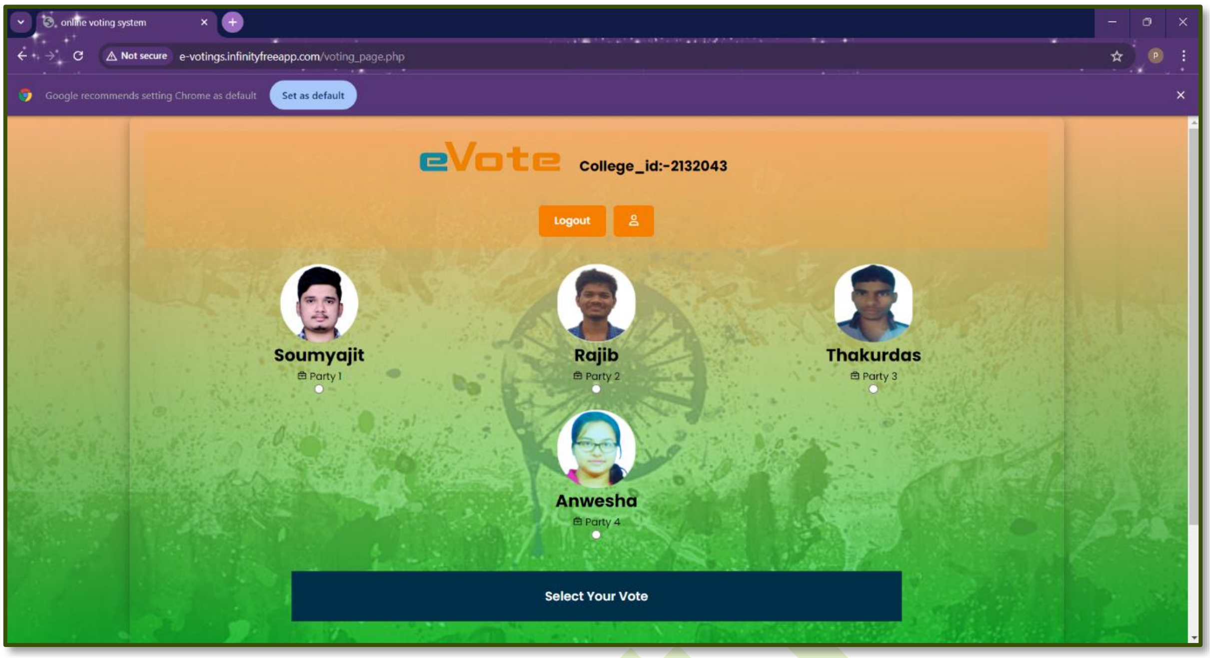
User Login:

After clicking on login button -



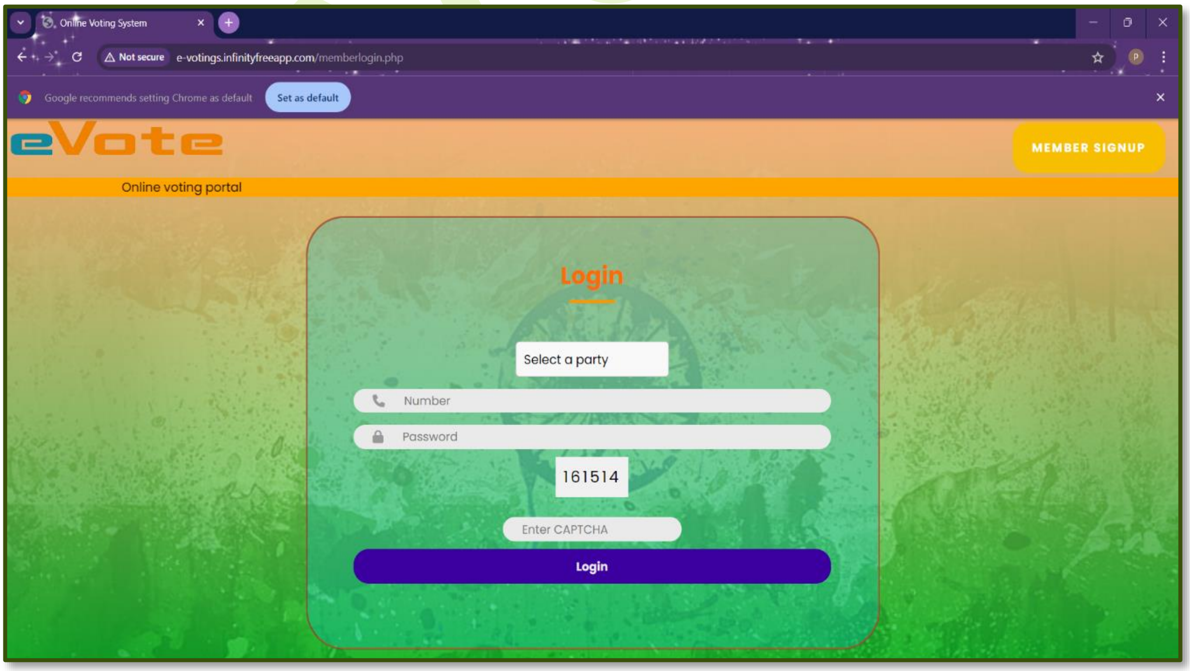
Voting Page:

After login, the voting page will be shown –

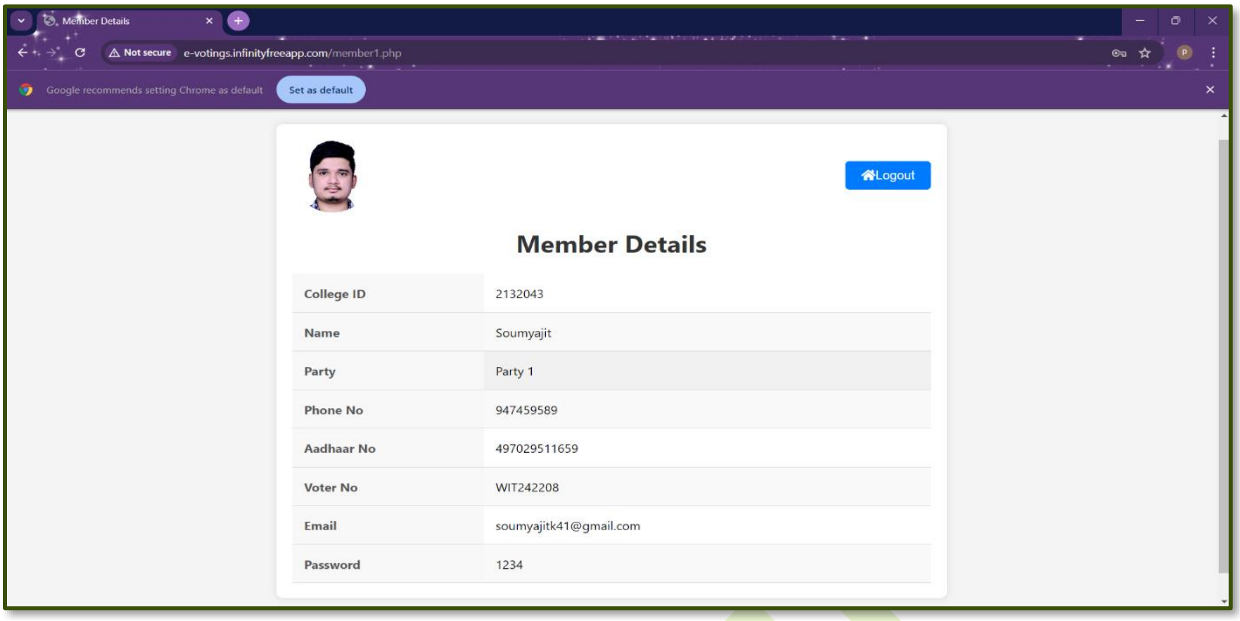


Member Login Page:

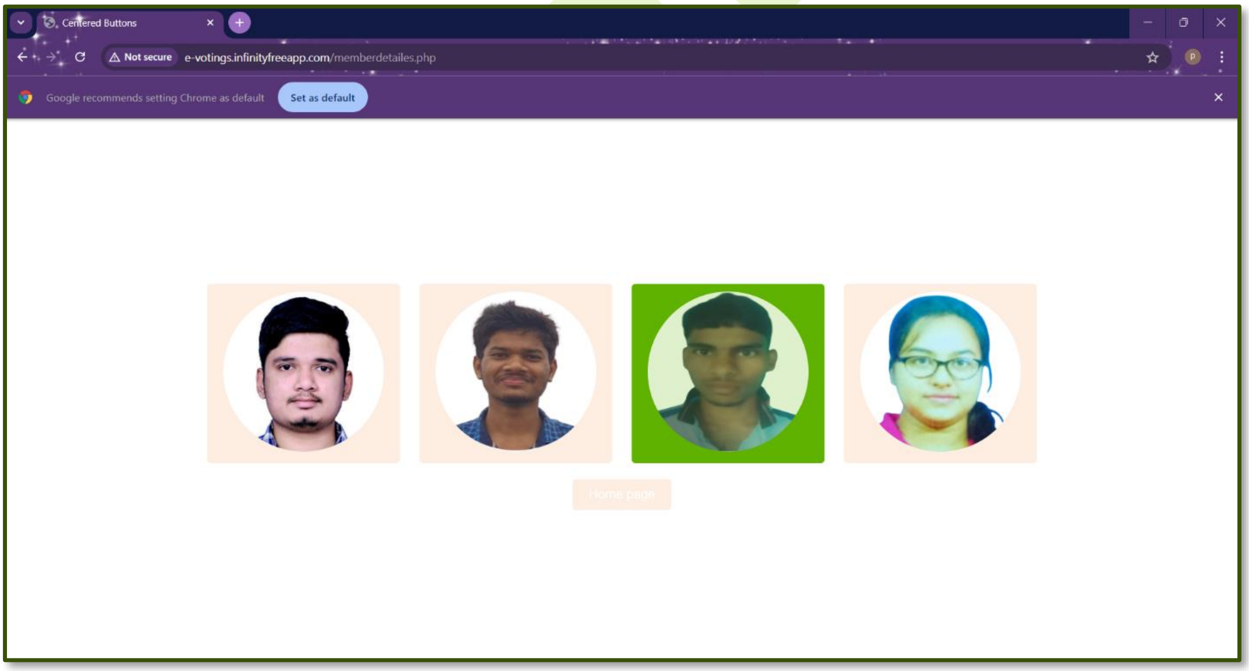
After clicking on the member login button –



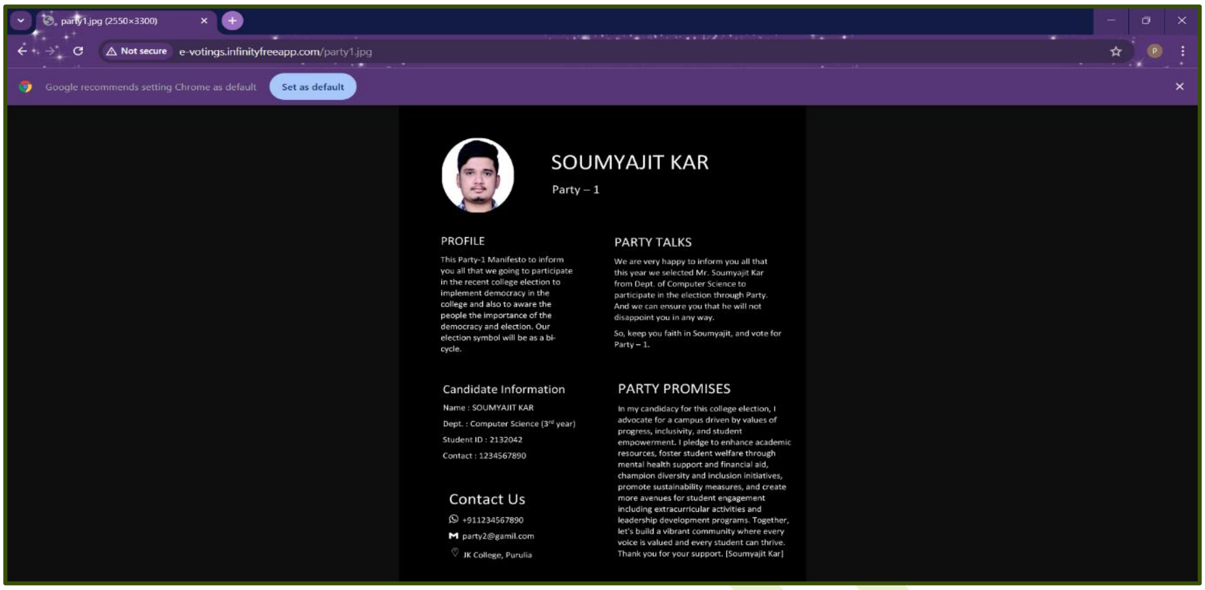
After member login this page will be shown -



View Member Page:

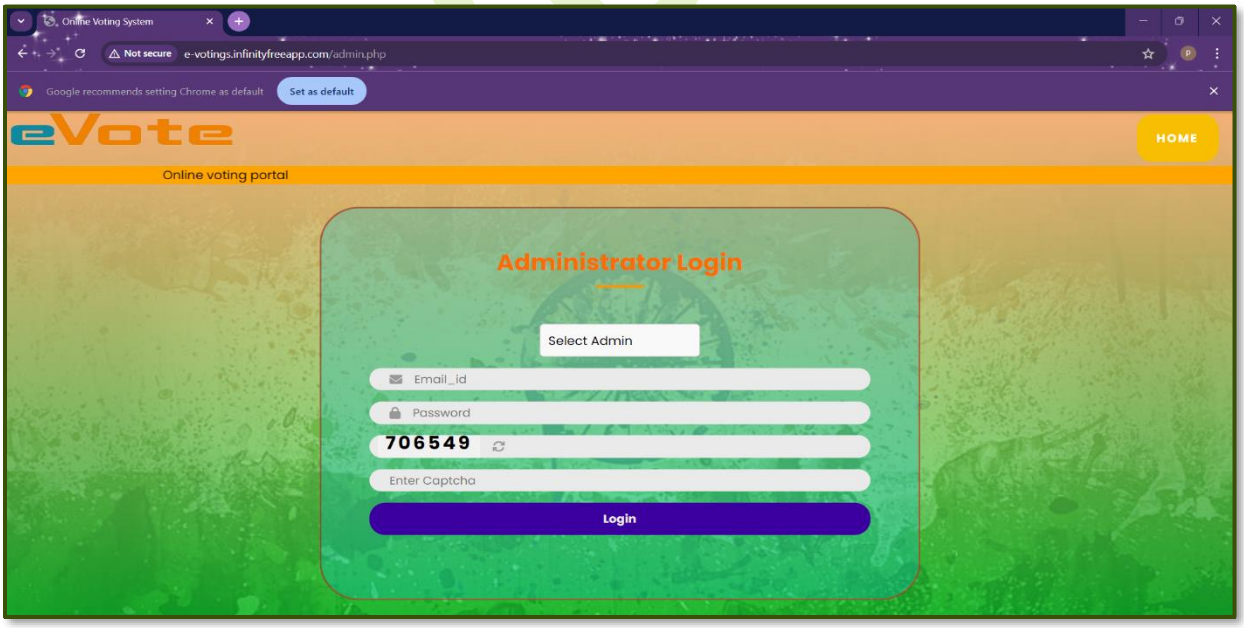


After clicking on any one member, it will show every information about party and member which is clicked.

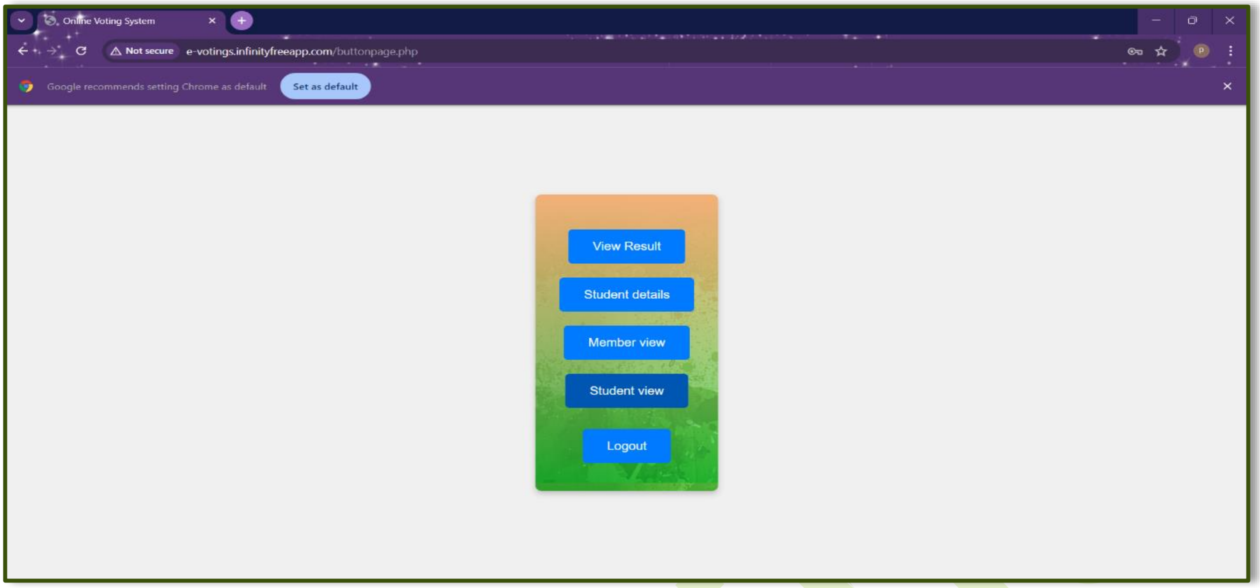


Administrator Page:

After clicking on the 'Admin Page' button –

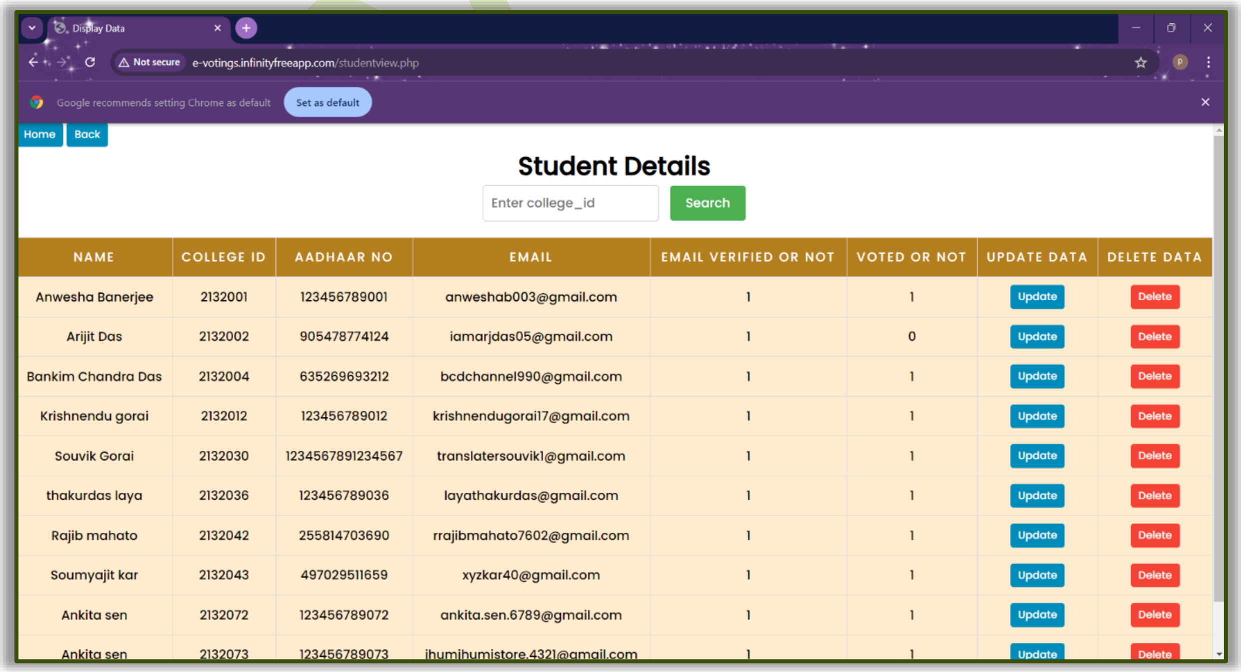


After login on administrator page, the below page will be shown –



In this page by clicking on different buttons admin can view result, view members, students who have registered and also can update. For example,

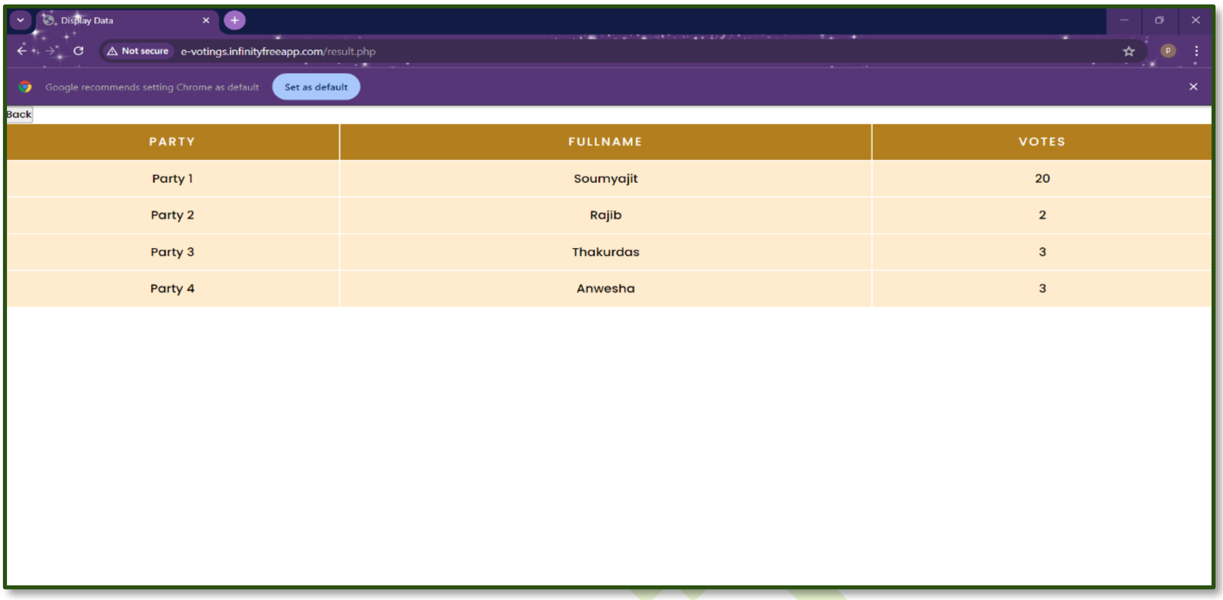
After clicking on student details button, this page will be shown –



NAME	COLLEGE ID	AADHAAR NO	EMAIL	EMAIL VERIFIED OR NOT	VOTED OR NOT	UPDATE DATA	DELETE DATA
Anwesha Banerjee	2132001	123456789001	anweshab003@gmail.com	1	1	Update	Delete
Arijit Das	2132002	905478774124	iamarjdas05@gmail.com	1	0	Update	Delete
Bankim Chandra Das	2132004	635269693212	bcdchannel990@gmail.com	1	1	Update	Delete
Krishnendu gorai	2132012	123456789012	krishnendugorai17@gmail.com	1	1	Update	Delete
Souvik Gorai	2132030	1234567891234567	translatersouvik1@gmail.com	1	1	Update	Delete
thakurdas laya	2132036	123456789036	layathakurdas@gmail.com	1	1	Update	Delete
Rajib mahato	2132042	255814703690	rrajibmahato7602@gmail.com	1	1	Update	Delete
Soumyajit kar	2132043	497029511659	xyzkar40@gmail.com	1	1	Update	Delete
Ankita sen	2132072	123456789072	ankita.sen.6789@gmail.com	1	1	Update	Delete
Ankita sen	2132073	123456789073	jhumjhumistore.4321@gmail.com	1	1	Update	Delete

Result Page:

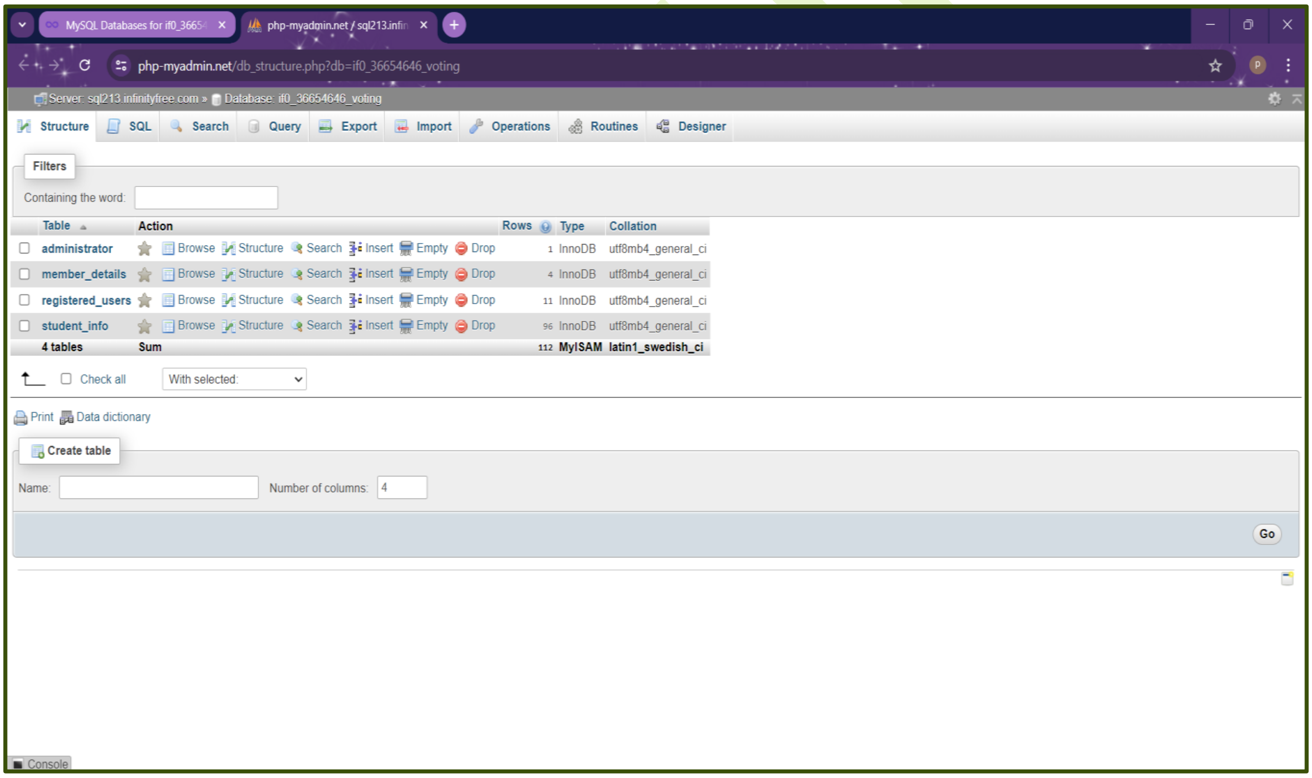
User can check the result by clicking on the ‘Check Result’ button in the home page –



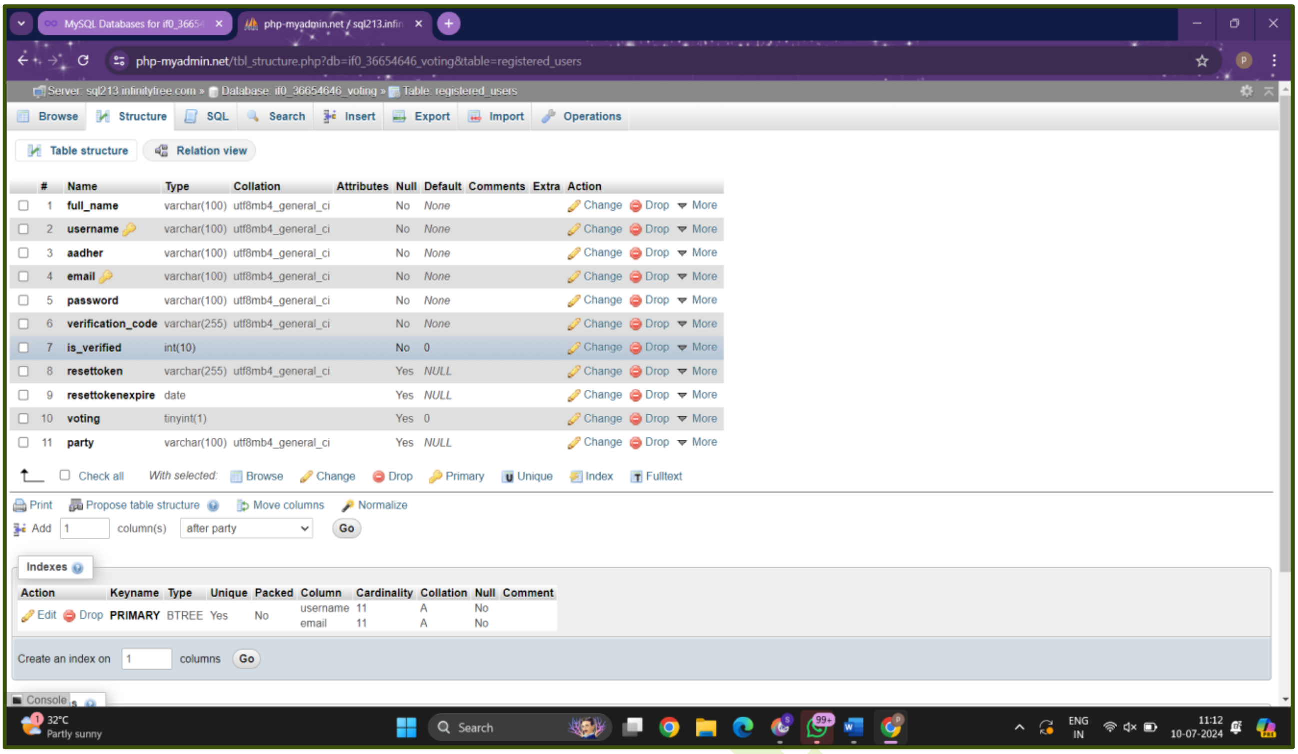
In this phase we have developed the tables, these are:

Table Design

ALL TABLE



STUDENT-DETAILS TABLE

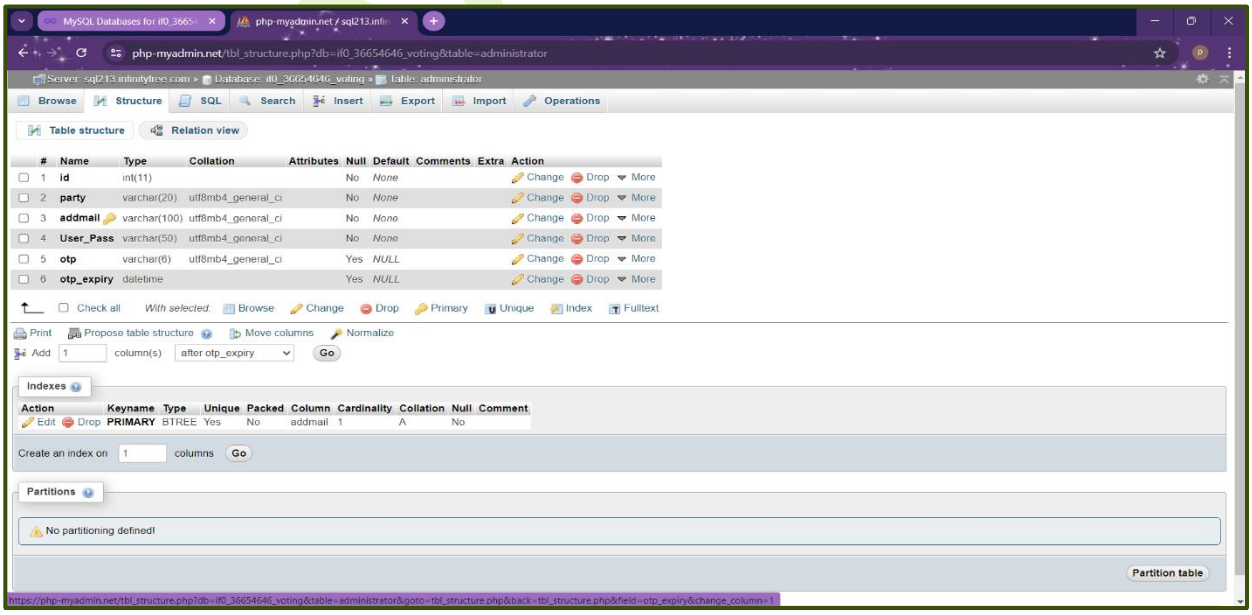


#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	full_name	varchar(100)	utf8mb4_general_ci		No	None			Change Drop More
2	username	varchar(100)	utf8mb4_general_ci		No	None			Change Drop More
3	aadher	varchar(100)	utf8mb4_general_ci		No	None			Change Drop More
4	email	varchar(100)	utf8mb4_general_ci		No	None			Change Drop More
5	password	varchar(100)	utf8mb4_general_ci		No	None			Change Drop More
6	verification_code	varchar(255)	utf8mb4_general_ci		No	None			Change Drop More
7	is_verified	int(10)			No	0			Change Drop More
8	resettoken	varchar(255)	utf8mb4_general_ci		Yes	NULL			Change Drop More
9	resettokenexpire	date			Yes	NULL			Change Drop More
10	voting	tinyint(1)			Yes	0			Change Drop More
11	party	varchar(100)	utf8mb4_general_ci		Yes	NULL			Change Drop More

Indexes

Action	Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
Edit Drop	PRIMARY	BTREE	Yes	No	username email	11 11	A A	No No	

MEMBER-DETAILS TABLE



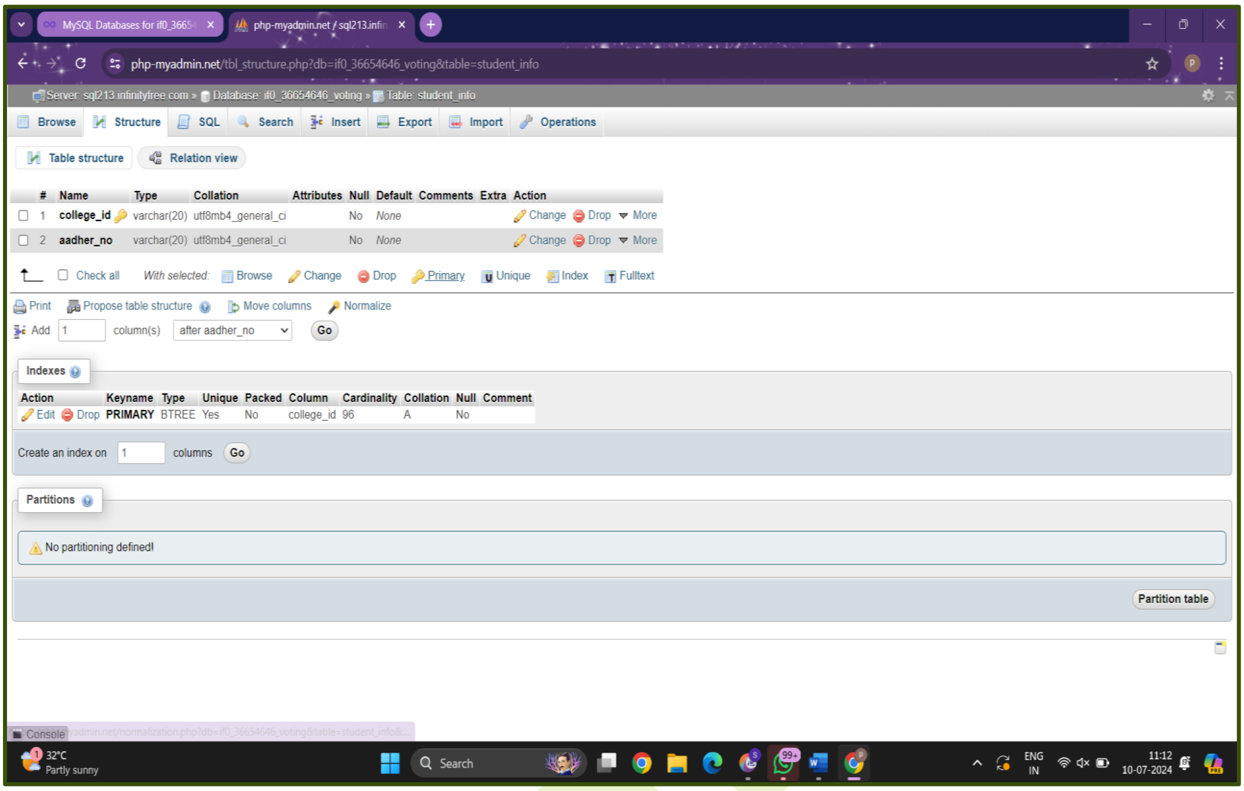
#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	id	int(11)			No	None			Change Drop More
2	party	varchar(20)	utf8mb4_general_ci		No	None			Change Drop More
3	addmail	varchar(100)	utf8mb4_general_ci		No	None			Change Drop More
4	User_Pass	varchar(50)	utf8mb4_general_ci		No	None			Change Drop More
5	otp	varchar(6)	utf8mb4_general_ci		Yes	NULL			Change Drop More
6	otp_expiry	datetime			Yes	NULL			Change Drop More

Indexes

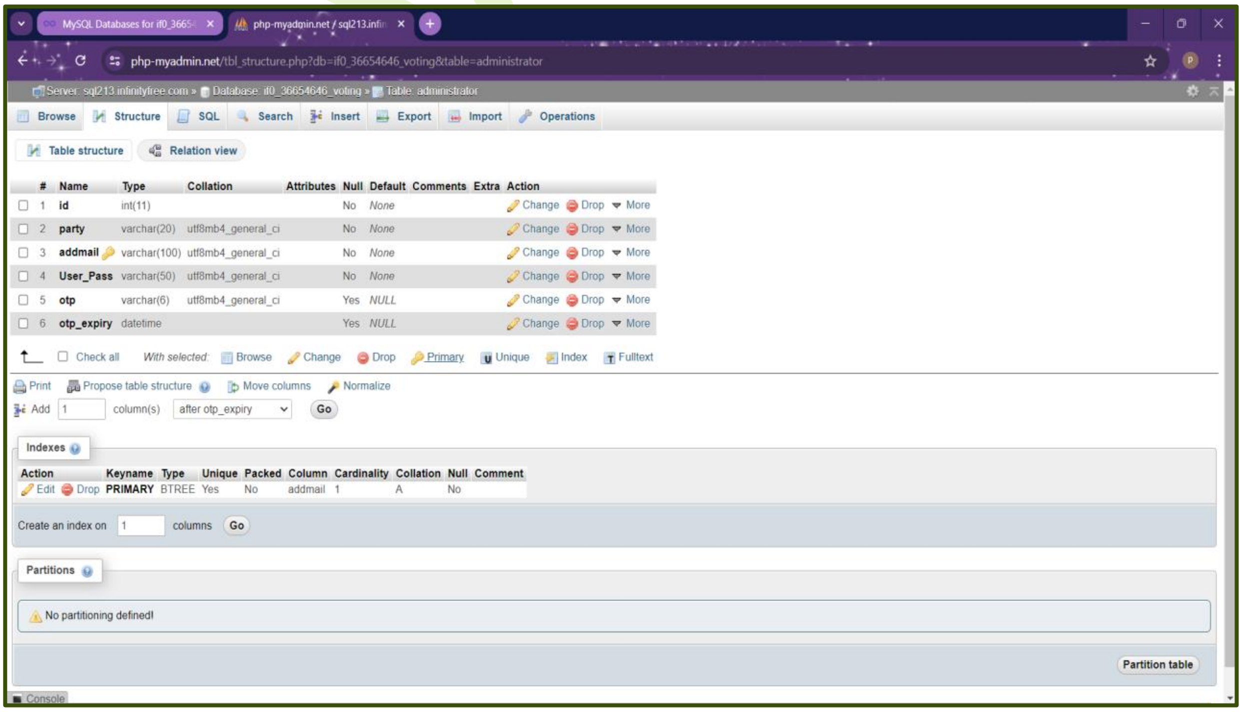
Action	Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
Edit Drop	PRIMARY	BTREE	Yes	No	addmail	1	A	No	

No partitioning defined!

COLLEGE DATABASE TABLE



ADMIN TABLES



TESTING

Testing on Online Voting System based on Agile model typically involves several iterative phases to ensure both functionality and security, these are as follows:

1. **Unit Testing:** In this phase we(developers) have tested individual components (like user interface, database interactions) to verify if they work as expected or not.
2. **Integration Testing:** Modules are combined and tested together to ensure that if they function correctly as a whole system or not.
3. **System Testing:** The entire voting system is tested end-to-end to verify that all components work together seamlessly or not.
4. **Security Testing:** Special attention is given to security measures such as encryption, authentication, and protection against hacking attempts.
5. **Performance Testing:** The system is tested under various loads to ensure that it performs well under typical voting conditions.
6. **Regression Testing:** After changes are made, previously completed tests are re-run to ensure existing functionality is not broken.

7. **User Acceptance Testing:** Final testing phase where stakeholders verify that the system meets their expectations and requirements.

These phases are typically repeated in short cycles(sprints) in Agile development, allowing for continuous improvement and adaptation based on feedback. Each phase is crucial to ensure the Online Voting System is reliable, secure, and user-friendly.

DEPLOYMENT

This project is deployed on infinity free app so that user can easily access it via link.

Deploying an online voting system involves several critical steps to ensure security, reliability, and accessibility. Here's an overview of the deployment process:

- Agile teams aim to release software frequently, typically at the end of each iteration or when a sufficient number of features have been completed and validated.
- Continuous deployment and delivery practices ensure that new features and updates are delivered to users as soon as they are ready and meet quality standards.

REVIEW

- Once the software is deployed, Agile teams monitor its performance and gather feedback from users and stakeholders.

- This feedback is used to inform future iterations, prioritize new features, and make adjustments to the product roadmap as necessary.

MAINTENANCE

Maintenance of Online Voting System using Agile model involves iterative cycles of development and maintenance. Here are the stages typically involved:

1. **Backlog Refinement:** Continuously refine the backlog to prioritize maintenance tasks such as bug fixes, security updates, and performance improvements.

2. **Sprint Planning:** Plan maintenance tasks for each sprint based on their priority and impact.

3. **Sprint Execution:** Execute maintenance tasks identified during sprint planning. This involves fixing bugs, addressing security vulnerabilities, and making necessary updates.

4. **Review and Retrospective:** At the end of each sprint, review completed maintenance tasks, gather feedback, and identify areas for improvement in maintenance processes.

By following these stages within an Agile model, we can effectively manage and maintain an Online voting System, ensuring it remains secure, reliable, and aligned with user expectations.

CONCLUSION

We all that how important voting is to select a well-deserved person for particular post. It implements transparency for giving everyone a fair chance. So, it is very necessary to conduct election in every case where one can control an administrative for a particular time period.

Through our web app (E-Voting) we are providing platform where the system doesn't need to worry about election time, space, fund etc. They just focus on their promises and on the manifesto.

This can also help as to conduct a paperless free and fair election which can improve our environmental status.

FUTURE SCOPE OF THE PROJECT

Future scope of this project seems very promising, with potential advancement of security measures, user authentication methods, accessibility features, and integration with emerging technologies like blockchain for increased transparency and trust. Some important scopes are below –

- ❖ Enhanced security feature with point-to-point data encryption and also using of some advance firewalls.

- ❖ Implementing biometric authentication such as fingerprint, face recognition it will provide an extra layer of security.
- ❖ With times improvement of the interface is needed. So that it can attract more user base with it.

Remote voting i.e. should be available for every location and in any time.

DRAWBACKS OF ONLINE VOTING SYSTEM

Online voting systems offer convenience and potential for increased participation, but they also come with several drawbacks:

1. Security Concerns: Online voting systems face significant security risks such as hacking, malware attacks, and potential manipulation of votes. Ensuring the integrity and confidentiality of votes is challenging.

2. Voter Authentication: Verifying the identity of voters online is difficult. Ensuring that each voter is eligible and only votes once while maintaining anonymity is a complex problem.

3. Digital Divide: Not everyone has equal access to the internet or technology, which could disenfranchise certain demographics, potentially leading to unequal participation.

5. Complexity and Costs: Developing and maintaining secure online voting platforms can be expensive and technically challenging. Ensuring accessibility for voters with disabilities or those who are not tech-savvy adds further complexity.

6. Potential for Coercion: Unlike traditional voting methods, online voting could be conducted under coercive conditions where someone is pressured to vote in a specific way, which may not be easily detectable.

REFERENCE AND BIBLIOGRAPHY

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- "Web Technologies" of dreamtech publication

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- InfinityFree